

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

# Exploratory Data Analysis Lab

## LAB ASSESSMENT 9

NAME : ARJUN MISHRA

REG NO : 23BDS0334

SLOT : L31+L32

COURSE CODE : BCSE331P

# Principal Component Analysis

1. **Load the dataset sent to your email into R, and verify that the packages ggplot2, ggfortify, FactoMineR, and factoextra are loaded.**

   **Aim:** To load the dataset into R and check that ggplot2, ggfortify, FactoMineR, and factoextra are available and loaded.

   **CODE:**

assessment_pca_9.R*

Source on Save

```r
#23BDS0334
#ArjunMishra
packages <- c("ggplot2", "ggfortify",
              "FactoMineR", "factoextra")

installed <- rownames(installed.packages())

for (p in packages) {
  if (!(p %in% installed)) {
    install.packages(p)
  }
}

# Load required libraries
library(ggplot2)
library(ggfortify)
library(FactoMineR)
library(factoextra)

# Load dataset
data <- read.csv("assessment.csv",
                 header = TRUE)

# Display first six rows
head(data)

# Verify packages
sapply(packages, requireNamespace,
       quietly = TRUE)
```

**OUTPUT:**

```
> #ArjunMishra
> packages <- c("ggplot2", "ggfortify",
+               "FactoMineR", "factoextra")
>
> installed <- rownames(installed.packages())
>
> for (p in packages) {
+   if (!(p %in% installed)) {
+     install.packages(p)
+   }
+ }
>
> # Load required libraries
> library(ggplot2)
> library(ggfortify)
> library(FactoMineR)
> library(factoextra)
>
> # Load dataset
> data <- read.csv("assessment.csv",
+                  header = TRUE)
>
> # Display first six rows
> head(data)
  rownames Gunnel Time Fromlow Slope  Rw Amphiso Subst Pool Water Cobble
1        1      0  786     190    10 0.1       0     3    0     0      0
2        2      0  794     182    10 0.0       0     3    0     0      0
3        3      0  798     178    20 0.0       0     1    0     0      0
4        4      0  800     176    70 0.3       0     1    0     1      0
5        5      0  804     172    40 0.3       0     1    0     1      0
6        6      0  808     168    60 0.3       0     1    0     1      0
>
> # Verify packages
> sapply(packages, requireNamespace,
+        quietly = TRUE)
   ggplot2  ggfortify FactoMineR factoextra
      TRUE       TRUE       TRUE       TRUE
```

**RESULT:**

The dataset is loaded into R, and the required packages are verified.

## 2. Display the total number of rows and columns in the dataset.

**AIM:** To determine the dimensions of the dataset.

**CODE:**

assessment_pca_9.R*

Source on Save

```
1  #23BDS0334
2  #ArjunMishra
3  # Display number of rows and columns
4  cat("Total rows:", nrow(data), "\n")
5  cat("Total columns:", ncol(data), "\n")
6
7  # Display dataset dimensions
8  dim(data)
```

**OUTPUT:**

R 4.5.2 · ~/

```
> #23BDS0334
> #ArjunMishra
> # Display number of rows and columns
> cat("Total rows:", nrow(data), "\n")
Total rows: 1592
> cat("Total columns:", ncol(data), "\n")
Total columns: 11
>
> # Display dataset dimensions
> dim(data)
[1] 1592   11
>
```

**RESULT:**

The dataset contains 1,592 rows and 11 columns.

3. **Extract only the numeric columns, clean the data, and normalize it using the scale() function into an object named data_scaled.**

**AIM:**

To select numeric variables, handle missing values, remove the identifier column, and normalize the data using scale().

**CODE:**

assessment_pca_9.R*

Source on Save

```r
#23BDS0334
#ArjunMishra
# Extract numeric columns
numeric_data <- data[
  , sapply(data, is.numeric),
  drop = FALSE
]

# Remove the identifier column
# 'rownames' is a record ID, not a PCA feature
numeric_data <- numeric_data[
  , names(numeric_data) != "rownames",
  drop = FALSE
]

# Remove duplicate column names if any
names(numeric_data) <- make.unique(names(numeric_data))

# Remove rows containing missing values
numeric_data <- na.omit(numeric_data)

# Remove zero-variance columns if any
valid_cols <- sapply(numeric_data, function(x) {
  sd(x) > 0
})

numeric_data <- numeric_data[, valid_cols, drop = FALSE]

# Normalize the data
data_scaled <- scale(numeric_data)

# Display results
cat("Rows after cleaning:", nrow(data_scaled), "\n")
cat("Columns after cleaning:", ncol(data_scaled), "\n")
head(data_scaled)
```

## OUTPUT:

```
> #23BDS0334
> #ArjunMishra
> # Extract numeric columns
> numeric_data <- data[
+   , sapply(data, is.numeric),
+   drop = FALSE
+ ]
>
> # Remove the identifier column
> # 'rownames' is a record ID, not a PCA feature
> numeric_data <- numeric_data[
+   , names(numeric_data) != "rownames",
+   drop = FALSE
+ ]
>
> # Remove duplicate column names if any
> names(numeric_data) <- make.unique(names(numeric_data))
>
> # Remove rows containing missing values
> numeric_data <- na.omit(numeric_data)
>
> # Remove zero-variance columns if any
> valid_cols <- sapply(numeric_data, function(x) {
+   sd(x) > 0
+ })
>
> numeric_data <- numeric_data[, valid_cols, drop = FALSE]
>
> # Normalize the data
> data_scaled <- scale(numeric_data)
>
> # Display results
> cat("Rows after cleaning:", nrow(data_scaled), "\n")
Rows after cleaning: 1592
> cat("Columns after cleaning:", ncol(data_scaled), "\n")
Columns after cleaning: 10
> head(data_scaled)
      Gunnel      Time   Fromlow      Slope         Rw    Amphiso       Subst       Pool      Water     Cobble
1 -0.1762627 0.5944359 0.4910469 0.03122942 -1.0137551 -0.5163247 -0.03785446 -0.5033727 -1.0174257 -0.4856713
2 -0.1762627 0.6533228 0.3942676 0.03122942 -1.2511280 -0.5163247 -0.03785446 -0.5033727 -1.0174257 -0.4856713
3 -0.1762627 0.6827663 0.3458780 0.66056149 -1.2511280 -0.5163247 -0.75528659 -0.5033727 -1.0174257 -0.4856713
4 -0.1762627 0.6974880 0.3216832 3.80722182 -0.5390094 -0.5163247 -0.75528659 -0.5033727  0.9822554 -0.4856713
5 -0.1762627 0.7269314 0.2732935 1.91922562 -0.5390094 -0.5163247 -0.75528659 -0.5033727  0.9822554 -0.4856713
6 -0.1762627 0.7563749 0.2249039 3.17788976 -0.5390094 -0.5163247 -0.75528659 -0.5033727  0.9822554 -0.4856713
>
```

## RESULT:

The numeric features are cleaned and normalized for PCA.

4. **Perform Principal Component Analysis on data_scaled using either the PCA() or prcomp() function, and print its summary.**

**AIM:** To reduce the dimensionality of the standardized dataset using Principal Component Analysis.

**CODE:**

assessment_pca_9.R*

Source on Save

```r
#23BDS0334
#ArjunMishra
# Perform PCA
pca_result <- prcomp(
  data_scaled,
  center = FALSE,
  scale. = FALSE
)

# Print PCA summary
summary(pca_result)
```

**OUTPUT:**

```
R 4.5.2 · ~/
> #23BDS0334
> #ArjunMishra
> # Perform PCA
> pca_result <- prcomp(
+   data_scaled,
+   center = FALSE,
+   scale. = FALSE
+ )
>
> # Print PCA summary
> summary(pca_result)
Importance of components:
                          PC1    PC2    PC3    PC4     PC5     PC6     PC7     PC8     PC9    PC10
Standard deviation     1.4924 1.3522 1.1673 1.0118 0.97663 0.91464 0.86517 0.71865 0.65062 0.28181
Proportion of Variance 0.2227 0.1828 0.1363 0.1024 0.09538 0.08366 0.07485 0.05165 0.04233 0.00794
Cumulative Proportion  0.2227 0.4056 0.5418 0.6442 0.73957 0.82323 0.89808 0.94973 0.99206 1.00000
>
```

**RESULT:**

PCA is performed successfully, producing principal components ordered by the amount of variance they explain.

5. **Plot a scree plot showing the percentage of variance for each component with data labels, setting the y-axis limit from 0 to 60.**

**AIM:** To visualize the percentage of variance explained by each principal component, with data labels and a y-axis limit from 0 to 60.

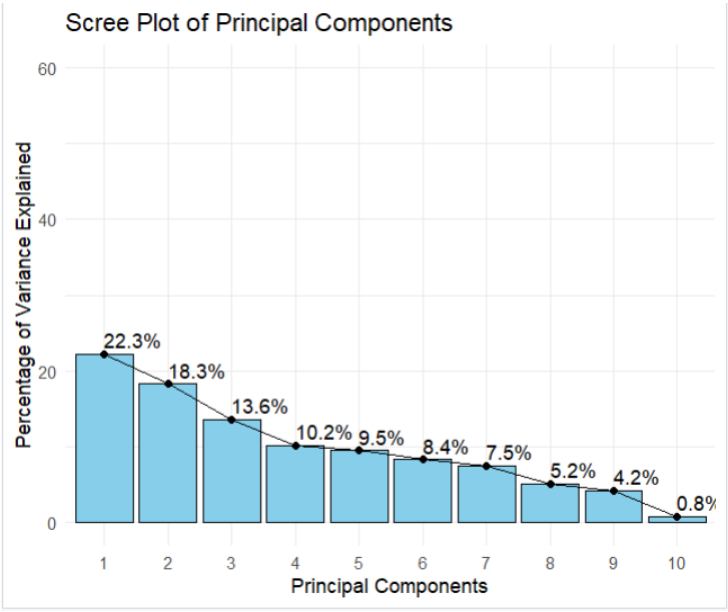
**CODE:**

assessment_pca_9.R* ×

Source on Save

```
#23BDS0334
#ArjunMishra
# Scree plot
fviz_eig(
  pca_result,
  addlabels = TRUE,
  ylim = c(0, 60),
  barfill = "skyblue",
  barcolor = "black"
) +
  labs(
    title = "Scree Plot of Principal Components",
    x = "Principal Components",
    y = "Percentage of Variance Explained"
  ) +
  theme_minimal()
```

**OUTPUT:**




**RESULT:**

The scree plot helps identify which principal components explain the greatest amount of variation in the dataset.

6. **Create a PCA biplot with non-overlapping labels and a proper title.**

**AIM:** To visualize observations and variable contributions in the principal component space, using non-overlapping variable labels.

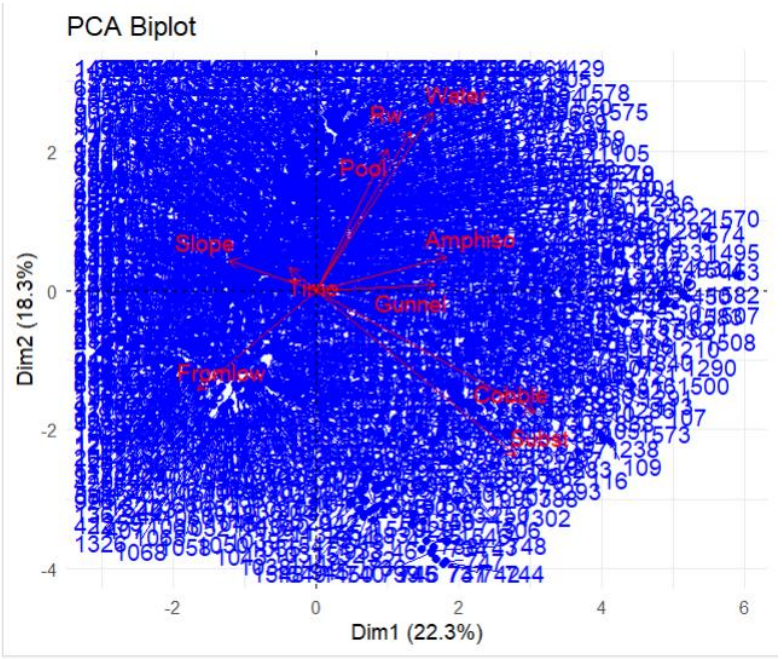
**CODE:**

assessment_pca_9.R*

Source on Save

```
#23BDS0334
#ArjunMishra
fviz_pca_biplot(
  pca_result,
  repel = TRUE,
  col.var = "red",
  col.ind = "blue",
  title = "PCA Biplot"
) +
  theme_minimal()
```

**OUTPUT:**




**RESULT:**

The biplot helps identify relationships between variables and the distribution of observations across the first two principal components.

7. **Extract the PCA scores into a data frame named pca_scores and display the first six rows.**

**AIM:** To extract the principal component scores into a data frame named pca_scores.

**CODE:**

assessment_pca_9.R*

Source on Save

```
#23BDS0334
#ArjunMishra
# Extract PCA scores
pca_scores <- as.data.frame(
  pca_result$x
)
# Display the first six rows
head(pca_scores, 6)
```

**OUTPUT:**

```
> #23BDS0334
> #ArjunMishra
> # Extract PCA scores
> pca_scores <- as.data.frame(
+   pca_result$x
+ )
> # Display the first six rows
> head(pca_scores, 6)
        PC1        PC2         PC3        PC4        PC5       PC6         PC7       PC8        PC9        PC10
1 -1.287426 -1.1060793 -0.31566532 -0.1857987 -0.1721037 0.2407359  0.38144193 0.4211681 0.03155594 -0.14202618
2 -1.317922 -1.1797523 -0.34461787 -0.2801779 -0.2136058 0.1960650  0.39966674 0.6095551 0.14138220 -0.10362253
3 -1.787989 -0.7862334 -0.09718681 -0.2954896  0.2834586 0.2630928  0.42082004 0.7752033 0.15758253  0.34020694
4 -1.724168  0.7719990  0.32813950 -0.2763229  3.2922112 0.4575543 -0.42674618 0.7007641 1.48602760 -0.12411631
5 -1.310470  0.6320196 -0.14916236 -0.1791404  1.6310810 0.2635427 -0.05977903 0.2991360 1.37882684  0.06371536
6 -1.566203  0.7505772  0.17107245 -0.2445225  2.7149752 0.4453315 -0.30432488 0.6309075 1.47158331 -0.05798533
>
```

**RESULT:**

The PCA scores are stored in pca_scores, and the first six observations are displayed.

8. **Draw a scatter plot of the first two principal components (PC1 vs. PC2) with blue circular points and your registration number in the title.**

**AIM:** To visualize the observations using the first two principal components with blue circular points and the registration number in the title.

**CODE:**

assessment_pca_9.R*

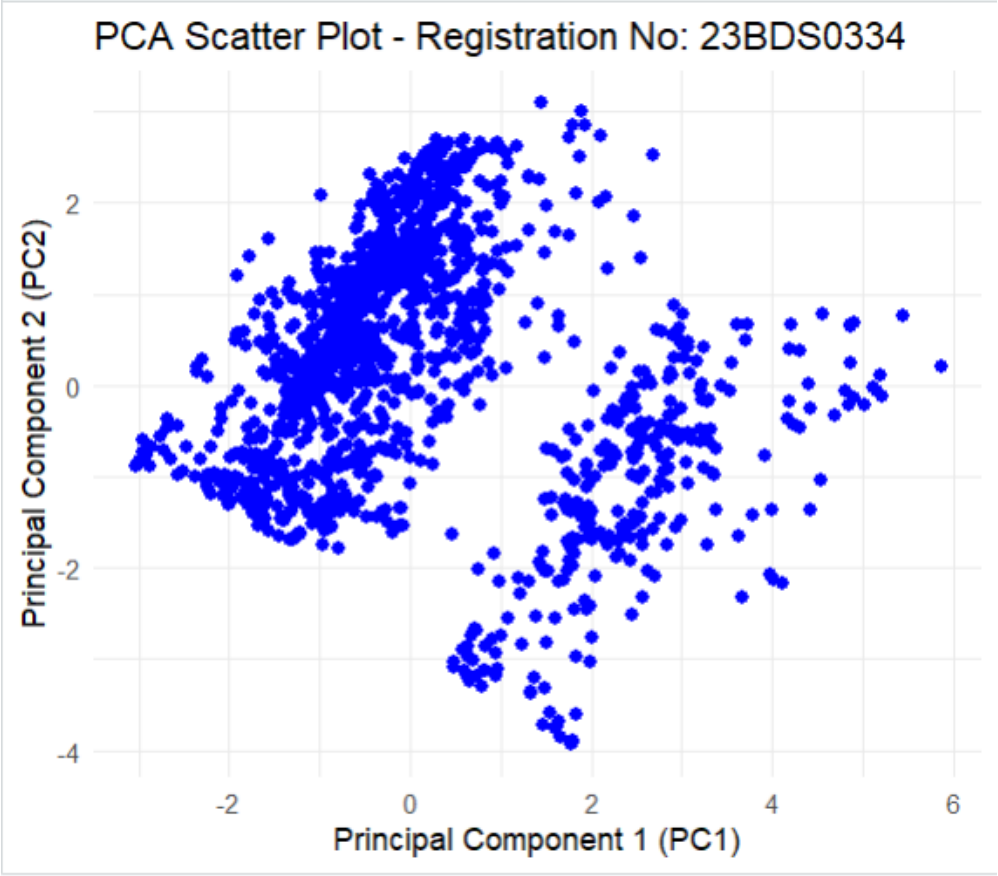
Source on Save

```
#23BDS0334
#ArjunMishra
reg_no <- "23BDS0334"

# Scatter plot of PC1 vs PC2
ggplot(pca_scores, aes(x = PC1, y = PC2)) +
  geom_point(
    color = "blue",
    shape = 16,
    size = 2
  ) +
  labs(
    title = paste(
      "PCA Scatter Plot - Registration No:",
      reg_no
    ),
    x = "Principal Component 1 (PC1)",
    y = "Principal Component 2 (PC2)"
  ) +
  theme_minimal()
```

**OUTPUT:**

```
> #23BDS0334
> #ArjunMishra
> reg_no <- "23BDS0334"
>
> # Scatter plot of PC1 vs PC2
> ggplot(pca_scores, aes(x = PC1, y = PC2)) +
+   geom_point(
+     color = "blue",
+     shape = 16,
+     size = 2
+   ) +
+   labs(
+     title = paste(
+       "PCA Scatter Plot - Registration No:",
+       reg_no
+     ),
+     x = "Principal Component 1 (PC1)",
+     y = "Principal Component 2 (PC2)"
+   ) +
+   theme_minimal()
>
```




**RESULT:**

The plot visualizes the distribution and possible clustering of observations in the first two principal component dimensions.